Dr. Himanshu Dave

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Education

- **Arizona State University** — Tempe, Arizona, USA
  Ph.D. Mechanical Engineering - (GPA - 3.64/4.00) — *May 2024*
  Advisors: Dr. Mohamed Houssem Kasbaoui (Chair), Dr. Marcus Herrmann (Co-Chair)
  Thesis title - Towards high fidelity multiphase simulations based on volume-filtering: from point-particle to interface resolved descriptions.
- **Arizona State University** — Tempe, Arizona, USA
  M.Sc. Mechanical Engineering - (GPA - 3.64/4.00) — *May 2023*
- **Arizona State University** — Tempe, Arizona, USA
  B.Sc. Mechanical Engineering (Honors) - (GPA - 3.25/4.00) — *May 2019*
  Honors thesis title - Design and analyze a liquid-liquid co-axial swirl injector for a small-scale rocket engine using CFD for minimum pressure drop and maximum spray angle.

Research Interests

• High-Fidelity Computational Fluid Dynamics (CFD) • Turbulence modeling • Compressible flows • Multiphase flows • Heat transfer within fluid flows

Fellowships and awards

- **National Science Foundation INTERN program** — Los Alamos, NM, USA
  Los Alamos National Laboratory — *Jul 2022 - Dec 2022*

Computing grants awarded

1. **XSEDE**: "Bridging the gap in multiphase flow simulations", PI: **Kasbaoui, M. H.**, Co-PI: Dave, H., 2M CPU hours, Period: 01/01/2021 to 12/31/2021.

Peer-reviewed Journal papers

3. **Dave, H.**, Brady, P., Herrmann, M. & Kasbaoui, M. H., Characterization of the forcing and sub-filter scale terms in the volume-filtering immersed boundary method, Journal of Computational Physics, 525, 113765 (03/15/2025). **(** 1)**

2. **Dave, H.** & Kasbaoui, M. H., Mechanisms of drag reduction by semi-dilute inertial particles in turbulent channel flow, Physical Review Fluids, 8, 084305 (08/21/2023). **(** 22)**

1. **Dave, H.**, Herrmann, M. & Kasbaoui, M. H., The volume-filtering immersed boundary method, Journal of Computational Physics, 487, 112136 (08/15/2023). **(** 11)**

**** Citation count from Google Scholar**

Peer-reviewed Conference papers

1. **Dave, H.** & Kasbaoui, M. H. Modulation of coherent structures by inertial particles in a turbulent channel flow, AIAA Scitech 2020 Forum, 1328 (01/05/2020).

Journal papers under review

1. **Dave, H.**, Patel, K., Zhu, X., Dynamics of melting interfaces under rotation, submitted to Journal of Fluid Mechanics.

Technical Skills

Computational Fluid Dynamics: Direct Numerical Simulation (DNS), Large-Eddy Simulation (LES), High-order unstructured solvers, Immersed Boundary Methods (IBM), Finite-volume methods, Flow-geometry coupling
Numerical Methods & Physics: High-order discretizations, stability analysis for hyperbolic PDEs, cut-cell methods, low-Reynolds number & compressible aerodynamics, turbulence modeling
High-Performance Computing: MPI parallelization, Linux clusters, scale-up optimization, exposure to GPU-accelerated CFD workflows
Programming & Automation: Python (advanced data analysis, scripting, and pipeline automation), C++, FORTRAN, MATLAB, Mathematica
Software & Solvers: CharLES, OpenFOAM, ANSYS Fluent, STAR-CCM+, Solidworks, Gmsh, VisIt, Paraview

Research Experience

- **Max Planck Institute (Computational Fluid Dynamics Group)** — Göttingen, Germany
  Postdoctoral Researcher — *September 2024 - Present*
  1. Conduct high-fidelity CFD simulations of turbulent flows with evolving complex interfaces using phase-field methods, characterizing boundary layer physics and heat transfer performance.
  2. Develop end-to-end Python simulation setup, automation, and post-processing pipelines to parse and extract key metrics from large-scale 3D datasets.
  3. Build memory-efficient visualization and data analysis tools for multi-terabyte HPC datasets, enabling rapid validation and design iteration of complex flow structures.
  4. Coordinate code deployment pipelines across high-performance environments to preserve software continuity.

- **Arizona State University** — Tempe, AZ, USA
  Graduate Teaching Assistant, School of Engineering of Matter — *January 2024 - May 2024*
  1. Instructed undergraduate students in thermal-fluid fundamentals, thermodynamics, and fluid dynamics principles to support experimental validation.
  2. Facilitated hands-on laboratory work for 100+ students on thermal-fluid cycles and aerodynamic characterization of systems.

- **Los Alamos National Laboratory** — Los Alamos, NM, USA
  Graduate Research Associate — *May 2023 - September 2023*
  1. Evaluated and optimized numerical stability of higher-order finite-differences for cut-cell methods using dispersive wave theory.
  2. Formulated novel high-order stable cut-cell stencils to solve hyperbolic conservation laws with complex geometric boundaries using automated scripts in Python and Mathematica.
  3. Implemented and scaled up high-order (up to 8th order) stencil algorithms in C++ within massive Linux HPC environments, reducing computational footprint while preserving accuracy.

- **Arizona State University** — Tempe, AZ, USA
  Graduate Research Associate (Advisor: Prof. M. H. Kasbaoui) — *Aug 2019 - May 2023*
  1. Developed a high-fidelity gas-solid multiphase solver in FORTRAN tailored for static and moving complex geometries using custom volume-filtering techniques.
  2. Benchmarked the custom code against canonical aeromechanics datasets, achieving 99.3% validation accuracy compared to experimental measurements.
  3. Extended the Volume-Filtered Immersed Boundary (VFIB) method to handle complex geometric boundaries that do not align with uniform structured meshes for acoustic and thermal analysis.
  4. Discovered a 20% skin-friction drag reduction mechanism in wall-bounded turbulence using advanced Eulerian-Lagrangian methods.
  5. Extracted structural flow dynamics from large datasets using spatial/temporal averaging to isolate performance-limiting vortical features.
  6. Managed code repositories using Git and utilized large-scale parallel scaling (OpenMPI) on Linux cluster resources to optimize numerical code throughput.

7. Mentored undergraduate students on high-performance computing (HPC) environments and data analysis scripting.

- **Los Alamos National Laboratory** — Los Alamos, NM, USA
  NSF Graduate Intern — *Jul 2022 - Dec 2022*
  1. Applied the specialized Volume-Filtered Immersed Boundary (VF-IB) framework to hyperbolic and parabolic PDEs to model thermal and acoustic environments near complex topological surfaces.
  2. Compared performance metrics against existing cut-cell solvers, proving identical resolution of flow physics with significantly lower grid-generation and computational costs.
  3. Quantified sub-grid scale (SGS) structural terms within filtering windows to assess grid-independence and model accuracy.

- **Arizona State University** — Tempe, AZ, USA
  Honors Thesis Project — *Aug 2018 - May 2019*
  1. Performed high-fidelity Large-Eddy Simulations (LES) of a liquid-liquid coaxial swirl injector under the high-order unstructured CharLES solver platform.
  2. Manufactured physical prototype hardware and validated aerodynamic/spray morphology metrics against LES data using experimental Particle Image Velocimetry (PIV) techniques.
  3. Managed thermal-structural optimization passes across the entire engine assembly to identify and eliminate high-flux thermal runoff scenarios.
  4. Co-founded a university rocketry organization, leading a 10-student propulsion engineering team through design, simulation, and hardware validation phases.

- **AzLoop (SpaceX Hyperloop Competition Team)** — Tempe, AZ, USA
  Team Lead - Braking, Stability & Aerodynamics — *Mar 2017 - Jul 2018*
  1. Conducted high-speed aerodynamic compressible flow analysis using CharLES and completed conjugate heat transfer runs in ANSYS Fluent to mitigate thermal failures.
  2. Executed multi-parameter vehicle stability and structural vibration studies using MATLAB to predict dynamic system limits.
  3. Managed precise CNC and lathe-based components fabrication processes to align exact parameters with computer modeling limits.

Conference Presentations & Proceedings

10. **Dave, H.**, Anas, M., Zhu, X., Melting within rotating Rayleigh-Bénard convection. Presented at: 2nd European Fluid Dynamics Conference; August 2025.

9. **Dave, H.**, Herrmann, M., Kasbaoui, M. H., Brady, P., A novel approach to solve PDE's involving boundary conditions on complex geometrical bounding surfaces using the Volume-Filtered Immersed Boundary (VF-IB) method. Presented at: 76th Annual Meeting of the APS Division of Fluid Dynamics; November 2023.

8. Kasbaoui M. H., **Dave H.**, Herrmann, M., A novel mass and momentum conserving immersed boundary method based on volume filtering. Presented at: 75th Annual Meeting of the APS Division of Fluid Dynamics; November 2022.

7. **Dave, H.**, Herrmann, M., Kasbaoui, M. H., A new conceptual approach for immersed boundaries based on volume-filtering. Presented at: 74th Annual Meeting of the APS Division of Fluid Dynamics; November 2021.

6. **Dave, H.**, Kasbaoui, M. H., Skin-friction drag modulation and riblet-like clusters in a semi-dilute particle-laden turbulent channel flow at $Re_\tau = 180$. Presented at: 74th Annual Meeting of the APS Division of Fluid Dynamics; November 2021.

5. **Dave, H.**, Kasbaoui, M. H., Modulation of Skin-friction Drag by Inertial Particles. In: 25th International Congress of Theoretical and Applied Mechanics. International Union of Theoretical and Applied Mechanics; 2021.

4. **Dave, H.**, Kasbaoui, M. H., A Novel Approach to Immersed Boundaries Based on the Volume-Filtering Framework. Presented at the: ASME 2021 Fluids Engineering Division Summer Meeting; August 2021.

3. **Dave, H.**, Kasbaoui, M. H., Turbulence modulation by inertial particles in Eulerian-Lagrangian simulations of a semi-dilute particle-laden channel flow. Presented at: 73rd Annual Meeting of the APS Division of Fluid Dynamics; November 2020.
2. **Dave, H.**, Kasbaoui, M. H., Modulation of coherent structures by inertial particles in a turbulent channel flow. Presented at: 72nd Annual Meeting of the APS Division of Fluid Dynamics; November 2019.
1. **Dave, H.** & Kasbaoui, M. H., Modulation of coherent structures by inertial particles in a turbulent channel flow, AIAA Scitech 2020 Forum, 1328 (01/05/2020).

Mentoring

1. **Joseph Crespo**: Fulton Undergraduate Research Initiative (FURI) student working on large domain particle-laden channel flows to evaluate size-scaling effects in semi-dilute regimes.
2. **Jack Madden**: Barrett Honors thesis student tracking how filtering windows and spatial discretization affect the calculated drag coefficient on spherical geometries using specialized IB approaches.

References

1. **Prof. M. Houssem Kasbaoui (PhD Advisor)**: houssem.kasbaoui@asu.edu
2. **Prof. M. Herrmann (PhD Co-Advisor)**: marcus.herrmann@asu.edu
3. **Prof. Xiaojue Zhu (Current Postdoc Advisor)**: zhux@mps.mpg.de